

Desktop AI-Powered Payroll Management System with Salary Prediction

Project Description: The Desktop AI-Powered Payroll Management System is a desktop application that automates payroll processing for organizations. It calculates employee salaries, manages tax deductions, overtime, bonuses, and generates salary slips. The system also uses Machine Learning to predict future salary trends based on employee performance, experience, attendance, and previous salary records.

Objectives

- Automate payroll calculations.
- Manage employee records.
- Calculate deductions (PF, Tax, ESI, Loan).
- Generate monthly salary slips.
- Predict future employee salaries using AI.
- Generate payroll reports.

Technologies

Python, Tkinter, SQLite, Scikit-learn, Pandas, NumPy, Matplotlib, VS Code/PyCharm.

Modules

Employee Management, Attendance, Payroll Processing, Deduction Management, AI Salary Prediction, Salary Slip Generation, Reports & Analytics, User Login.

Database Tables

Users, Employees, Attendance, Payroll, Deductions, Salary Prediction, Salary History.

AI Model

Algorithm: Linear Regression.

Inputs: Basic Salary, Years of Experience, Attendance Percentage, Performance Rating, Overtime Hours.

Output: Predicted Future Salary.

Requirements

Windows 10/11, Python 3.11+, SQLite, Intel Core i3+, 4GB RAM, 500MB Storage.

Expected Output

Fast payroll processing, accurate salary calculation, reduced manual errors, future salary prediction, professional reports.

Future Enhancements

Cloud DB, Facial Recognition, Email Salary Slips, Mobile App, Biometric Attendance.

Folder Structure

Payroll_System/: main.py, login.py, employee.py, payroll.py, attendance.py, prediction.py, database.py, reports.py, requirements.txt, employees.db, salary_slips/, reports/, models/, assets/.

Sample Salary

Basic ₹35,000; HRA ₹7,000; DA ₹3,500; Overtime ₹2,000; Gross ₹47,500; PF ₹2,100; Tax ₹1,500; ESI ₹500; Net ₹43,400.

Conclusion

Provides efficient payroll automation with AI salary prediction suitable for

MCA/BCA/B.Tech projects.